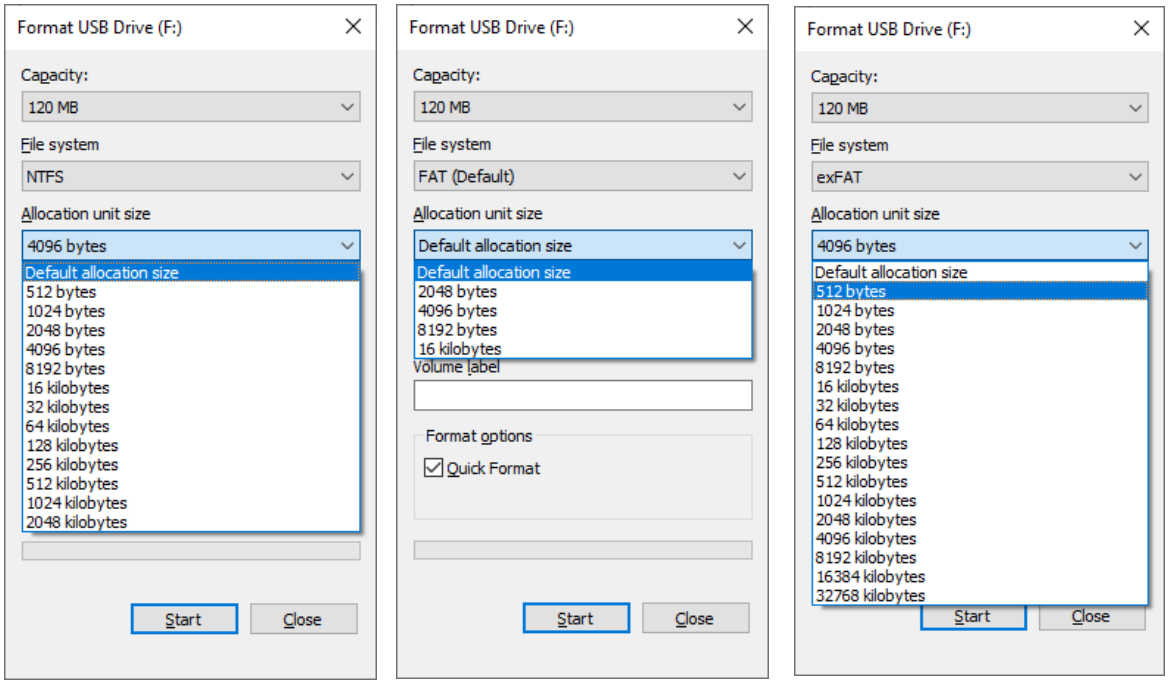
Unallocated Space & Slack Space

This additional discussion suplements lab 2 where these concepts are briefly covered.

Sectors, Blocks, Clusters and Allocation Units

Recall that when a storage medium is formatted, the medium (hard drive or USB stick for example) is divided into sectors and clusters (clusters are also called blocks). The sectors are usually 512 bytes but clusters may be a facor of this with 4096 bytes being common. When formatting the drive, the user will usually get an opportunity to select the number of sectors that make up a cluster (clusters are called allocation units in NTFS). This must either be 1 sector per cluster or multiples of 2 sectors per cluster, so 512bytes, and then in the case of FAT, 2048 (4 sectors) 4096 (8 sectors) etc, and in the case of exFAT, up to 32768 kilobytes.



In general, smaller cluster sizes store information more efficiently but are slower to write to, particularly if large files are saved. Therefore, some thought should be used when selecting cluster sizes – small files, small cluster size, large files, larger cluster sizes.

The example below shows what happens when a USB stick is formatted with NTFS and a cluster size of 2048 kilobytes. A small file of 11,937 bytes is created in Notepad and saved onto the stick.

Note, that a kilobyte is 1000 bytes and a kibibyte is 1024 bytes. The stick was formatted with a cluster size (cluster or allocation unit) of 2048 kilobytes, yet the calculation from the “size on disk’ of the file below shows 2MB (megabytes) and 2,097,152 bytes is reserved for the file (allocated). It is clear that in this instance, a kilobyte is intended to mean 1024 bytes and a megabyte, 1024 kilobytes. The confusion over terms remains, even in Windows 10 and 11.

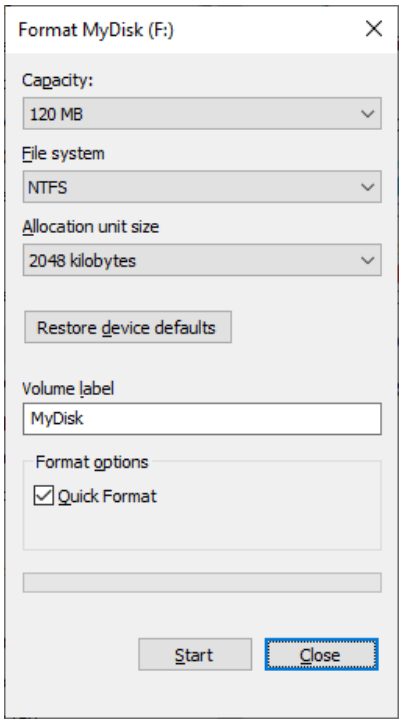
2048 x 1024 = 2,097,152

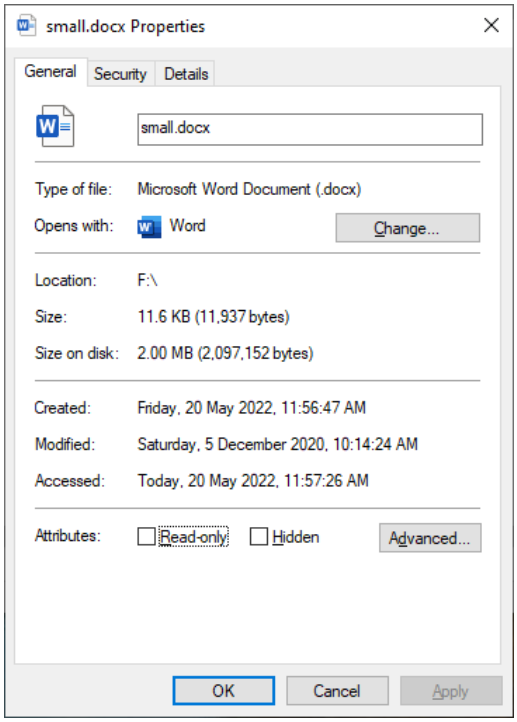
A file of 11,937 bytes has 1 cluster totalling 2,097,152 bytes allocated to it. The file only occupies less that 0.6% of the cluster, yet the whole cluster is now unavailable (allocated) for storage of other files. The space is marked as 'allocated'. A calculation of the 'size on disk' reserved for the file, minus the actual file size, shows less that 0.6% of the cluster is actually needed for this file.

2,097,152 – 11,937 = 2,085,215 bytes

(100 / 2,097,152) * 2085215 = 99.43% unused but unavailable.

Therefore, there is over 2 million bytes of slack space in this cluster that could be used to hide another file if the file is manually 'placed' in this area without the operating system assistance.





Note that if many files of this size (millions) are saved, the hard drive will begin to fill up quite quickly, yet almost 99.4% of the hard drive will not be used. Large cluster sizes for disks used to save small files is very inefficient.

Unallocated Space:

When space on a hard drive is written to, such as when a file is saved, the operating system will prevent other files that are saved being written over the file. The process for saving a file involves the operating system requesting the hard drive to save a file, the hard drive checks the file system information and looks for space on the hard drive that is not currently used. This is called unallocated space. The hard drive then saves the file onto this space and marks the space in the FAT (File Allocation Table for FAT file systems) or MFT (Master File Table for NTFS file systems) as allocated. Allocated space cannot be written to by a new file.

When the file is deleted, the file system makes a note that the space that is occupied by the file is now unallocated and free to be written to by another file if it is saved. The note in the file system, in the case of FAT file systems, is simply made by changing the start of the file location information in the FAT to E5 and wiping the clutser chain information with zeros. This is how forensic software recovers a file: it looks for file signatures or it looks in the FAT or MFT for E5 and locates the clusters where the deleted file is. Forensic Software sometimes has an option to only look for deleted files and/or files in slack space and by examing the FAT or MFT, these files can be located. That is. until a new file is written to the same location.

Slack Space:
Slack space refers to a number of different areas of unusable space, as opposed to unused space such as unallocated space. Slack space is not available for file storage.

Volume slack space: If a drive is divided into partitions and one of those partitions is not formatted, then files cannot be saved to that partition. In the example, Partition 3 is volume slack space.

Partition 1 NTFS	Partition 2 NTFS	Partition 3 Not Formatted

File system slack space: Recall that a cluster (allocation unit) is a multiple of sectors. If, at the end of a file system, there are not enough sectors to constitute a cluster, then those sectors cannot be written to. These sectors will remain unavailable and are called system slack space.

Example with 4 sectors per cluster – the final 2 sectors are system slack space

Sector 1	Sector 2	Sector 3	Sector 4	Sector 1	Sector 2

File slack RAM: Not to be confused with RAM memory, RAM file system slack space is from the last byte of a file in a sector until the end of the sector. In the example below, this is final, unused part of sector 2.

Sector 1	Sector 2		Sector 3	Sector 4

Drive slack space: This is the slack space from the last byte of the file until the end of the cluster. In the example above, this is the final part of sector 2 and sectors 3 and 4.

It is important to understand these terms and their implications for the forensic investigator. Ensure you understand how slack space can be used by a skilled person to hide files.